External Merge Sort for Large Employee Records Project

Project Overview

This project aims to implement the External Merge Sort algorithm to sort a large dataset of employee records stored in multiple files. The dataset will consist of 16 files, each containing 1,000 employee records, for a total of 16,000 records.

External Merge Sort

External Merge Sort is a sorting technique designed for large datasets that cannot fit entirely in memory. It works by splitting the data into smaller, manageable chunks, sorting them individually (using an in-memory sorting algorithm), and then merging the sorted chunks into a single, larger sorted file. This approach minimizes memory usage while maintaining efficient sorting performance.

Project Objectives

- Implement the External Merge Sort algorithm in a chosen programming language (e.g., Python, Java, C++).
- Design the data structure for employee records, including relevant fields (e.g., employee ID, name, department, salary).
- Generate 16 random files containing 1,000 employee records each.
- Implement functions to:
    - Read and write employee record files.
    - Sort individual files using an in-memory sorting algorithm.
    - Merge sorted files in a multi-way fashion (handling multiple input files simultaneously).
    - Handle potential overflows during the merge process (when the combined data exceeds available memory).
- Measure the time complexity of the sorting and merging operations.
- Evaluate the effectiveness of External Merge Sort for sorting large datasets.

Employee Record Structure:

The employee record structure will be defined within the program. Each record could include fields like:

- Employee ID (integer)
- Last Name (string)
- First Name (string)
- Department (string)
- Salary (float)

Sorting Criterion:

The project will focus on sorting the employee records based on a single criterion. The chosen criterion could be:

- Employee ID (ascending order): This is a common choice for efficient retrieval by employee ID.
- Last Name (ascending order): Useful for generating alphabetical employee lists.

Deliverables

- A complete, well-documented, and functional program implementing External Merge Sort.
- A report summarizing the project including:
  - Algorithm selection rationale and implementation details.
  - Data structure design choices.
  - Explanation of random data generation and file handling.
  - Time complexity analysis for sorting and merging operations.
  - Discussion of the results and evaluation of External Merge Sort's performance.

Evaluation Criteria

The project will be evaluated based on:

- Functionality: Correctness of the implemented algorithm and successful sorting of the data.
- Code quality: Readability, modularity, and proper documentation.
- Efficiency: Time complexity analysis and demonstration of efficient sorting performance.
- Report quality: Clarity, organization, and insightful discussion of the project outcomes.

Project Deadline

Sunday 25th May 2025